

IDS-SDD-JLR: IDS Symptom Driven Diagnostics



www.jlridssddmongoose.com

This publication is intended for instructional purposes only. Always refer to the appropriate service publication for specific details and procedures.

All rights reserved. All material contained herein is based on the latest information available at the time of publication. The right is reserved to make changes at any time without notice.

© 2012 Jaguar Land Rover North America LLC

<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

IDS-SDD-JLR: IDS Symptom Driven Diagnostics



General Information



www.jlridssddmongoose.com

This publication is intended for instructional purposes only. Always refer to the appropriate service publication for specific details and procedures.

All rights reserved. All material contained herein is based on the latest information available at the time of publication. The right is reserved to make changes at any time without notice.

© 2012 Jaguar Land Rover North America LLC

<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

Introduction	2
Acronyms and Abbreviations	3
Overview	7
IDS Specifications	8
IDS Software	9

Objectives

At the end of this course, technicians will be able to:

- Recognize IDS hardware
- Navigate through SDD menus
- Perform diagnostics in a logical sequence
- Diagnose the root cause of stored DTCs
- Utilize Pinpoint Tests
- Monitor/control systems using Datalogger

Online Course Evaluation

Class participants are encouraged to fill out an online evaluation for this course.

The Jaguar evaluation is available at:

- <http://www.hostedsurvey.com/takesurvey.asp?c=JLRUSJAG1>

The Land Rover evaluation is available at:

- <http://www.hostedsurvey.com/takesurvey.asp?c=JLRUSLR2>

The information provided in the evaluations is kept confidential and will only be used to improve training activities.

Your prompt response will be appreciated.

Your feedback is extremely important to us.

ACRONYMS AND ABBREVIATIONS

The following acronyms, abbreviations and symbols are used in this course book. The majority of them conform to J1930 standards.

Abbreviation	Definition
CCF	Car Configuration File
CD	Compact Disc
CDROM	Compact Disc Read Only Memory
CD-RW	Compact Disc Read/Write
CM	Control Module
CPU	Central Processing Unit
DDR	Double Data Rate
DDW	Dealer Direct Warranty
DLC	Data Link Connector
DMM	Digital Multimeter
DSG	Digital Signal Grouping
DTC	Diagnostic Trouble Code
DTS	Designated Technical Specialist
DVD	Digital Versatile Disc
FRFT	First Right First Time
GB	GigaByte
GHz	GigaHertz
GM MRDB	Global Master Reference Database
HHT	Hand-Held Tester
IDS	Integrated Diagnostic System
IT	Information Technology
LAN	Local Area Network
LCD	Liquid Crystal Display
LCS	Land Rover Coding System
LED	Light-Emitting Diode
LTFT	Long Term Fuel Trim
MB	MegaByte

Abbreviation	Definition
MRDB	Master Reference Database
MY	Model Year
OBD	On Board Diagnostics
OSC	Output State Control
PCMCIA	Personal Computer Memory Card International Association
PDI	Pre-Delivery Inspection
PWM	Pulse-Width Modulated
RAM	Random Access Memory
RDS	Land Rover Diagnostic System
ROM	Read Only Memory
SAE	Society of Automotive Engineers
SD	Secure Digital
SDD	Symptom-Driven Diagnostics
SDRAM	Synchronous Dynamic Random Access Memory
SMS	Support Management System
SSM	Special Service Message
TOPIx	Technical Online Product Information Exchange
TSB	Technical Service Bulletin
TSD	Touch Screen Display
T4	T4 Diagnostic System
UPS	Update Prior to Sale Notice
USB	Universal Serial Bus
VCI	Vehicle Communication Interface
VCM	Vehicle Communication Module
VMM	Vehicle Measurement Module
VIN	Vehicle Identification Number
VVA	Vehicle Vibration Analyzer

Acronym Cross-Reference

The following table cross-references Master Reference Data Base (MRDB) terms with some more familiar alternate terms. The MRDB terms are used by SDD to comply with the ISO 14229 standard for Unified Diagnostic Services.

MRDB Acronym	MRDB Description	Alternate Name(s)
AAM	Audio Amplifier Module	Audio Amplifier (AUD)
AAMB	Audio Amplifier Module B	Subwoofer Module
ABCM	Auxiliary Body Control Module	
ABS	Anti-Lock Brake System (ABS) Control Module	Anti-Lock Brake Module
ACCM	Air Conditioning Control Module	
ACM	Audio Front Control Module	Audio Module, Integrated Audio Module, IAM 2.1
ANT	Antenna Control Module	Antenna Module
APIM	Accessory Protocol Interface Module	Accessory/Auxiliary USB Unit
ARCM	Active Roll Control Module	
ATCM	All Terrain Control Module	Terrain Optimization Switch, Terrain Response Control Module
AVCM	Active Vibration Control Module	
AWD	All Wheel Drive Module	
BCBM	Body Control Module B	Rear Smart Junction Box, RJB
BCM	Body Control Module	Central Electronic Smart Junction Box, CJB
BECM	Battery Energy Control Module	Battery Management System Module
CCM	Cruise Control Module	Adaptive Cruise Control, Forward Sensing Module
CDP	Compact Disc Player / Changer Module	
CHCM	Chassis Control Module	
CHCMB	Chassis Control Module B	
CM	Compass Module	
CMR	Camera Module – Rear	Park Assist Camera
CTCM	Coolant Temperature Control Module	
DACMC	Digital Audio Control Module C	High Definition Radio
DDM	Driver Front Door Module	Driver Door Module
DPM	DVD Player / Changer Module	DVD Player
DRDM	Driver Rear Door Module	Rear Driver Door Module, LH Rear Door Module
DSM	Driver Front Seat Module	Seat Control Module A, Power Seat Module
DSP	Audio Digital Signal Processing Module	
EMM	Energy Management Module	
FACM	Fuel Additive Control Module	
FCDIM	Front Control/Display Interface Module	Combined HLDF and ICM, Touch-Screen Display (TSD)
FCIM	Front Controls Interface Module	
FCM	Fuel Cell Control Module	
FDM	Front Display Interface Module	

MRDB Acronym	MRDB Description	Alternate Name(s)
FDSM	Front Distance Sensing Module	Closing Velocity Module
FEM	Front Entertainment Control Module	
FLM	Front Lighting Control Module	
FPAM	Front Parking Assist Control Module	
GPSM	Global Positioning System Module	
GSM	Gear Shift Module	Electronic Transmission Selector (ETS)
GSMB	Gear Shift Module B	
GWM	Gateway Module	
HCM	Headlamp Control Module A	AFS Headlamp Control Module, Dynamic Headlamp Control
HCMB	Headlamp Control Module B	Auto Headlamp Dipping (AHD)
HUD	Head Up Display	
HVAC	HVAC Control Module	Climate Control Module
ICM	Information Center Module	Infotainment Control Module
ICS	Impact Classification System	
ILCM	Interior Lighting Control Module	
IPC	Instrument Panel Cluster (IPC) Control Module	Driver Information Module, Instrument Cluster
IPMA	Image Processing Module A	Night Vision System
IPMB	Image Processing Module B	Parking Multi Camera System
IRVM	Inside Rearview Mirror Module	
NCM	Navigation Control Module	MMM Multimedia Module
OCSS	Occupants Restraints Occupant Classification System Module	Occupant Weight Sensor
PBM	Park Brake Control Module	
PCM	Powertrain Control Module	Engine Control Module (ECM)
PDM	Passenger Front Door Module	Passenger Door Module
PMM	Powertrain Control Monitor Module	
PRDM	Passenger Rear Door Module	Rear Passenger Door Module, RH Rear Door Module
PSCM	Power Steering Control Module	Electric Power Assisted Steering Module
PSM	Passenger Front Seat Module	Seat Control Module B
RACM	Rear Audio Control Module	Rear Seat Entertainment (RSE) Module
RBM	Running Board Control Module	
RCM	Restraints Control Module	Supplementary Restraint System Module
RDCM	Rear Differential Control Module	DEM Differential Electronic Module, E-Coupling
RDUM	Remote Driver Utility Module	Remote Accessory Module
REM	Rear Entertainment Control Module	Rear Seat Entertainment (RSE)
RFA	Remote Function Actuator	Keyless Vehicle Module (KVM)
RGTM	Rear Gate / Trunk Module	Power Operated Tailgate
RHVAC	Rear HVAC Control Module	Rear Climate Control Module

MRDB Acronym	MRDB Description	Alternate Name(s)
RLM	Ride Level Control Module	
RSM	Remote Start Module	
SARM	Satellite Radio Module	Satellite Digital Audio Receiver (SDAR) Module, Remote Digital Audio Receiver (RDAR)
SASM	Steering Angle Sensor Module	Steering Angle Sensor
SCCM	Steering Column Control Module	Steering Wheel Module
SCMC	Seat Control Module C	Driver Rear Seat Module
SCMD	Seat Control Module D	Passenger Rear Seat Module (or power fold)
SCME	Seat Control Module E	Drivers Climate Seat, Seat Heating Module
SCMF	Seat Control Module F	Passenger Climate Seat, Rear Climate Seat
SCMG	Seat Control Module G	Drivers Seat Contour Control
SCMH	Seat Control Module H	Passenger Seat Contour Control
SECM	Steering Effort Control Module	
SODL	Side Object Detection Control Module – Left	Blind Spot Monitoring System Left
SODR	Side Object Detection Control Module – Right	Blind Spot Monitoring System Right
SPCM	Secondary Powertrain Control Module	
SPMA	Seatbelt Pretensioner Control Module A	
SPMB	Seatbelt Pretensioner Control Module B	
SPRM	Speech Recognition Module	Speech Recog. Module
SUMB	Suspension Control Module B	Adaptive Damping Control Module
TACM	Throttle Actuator Control Module	
TBM	Tow Bar Control Module	
TCCM	Transfer Case Control Module	
TCM	Transmission Control Module	
TU	Telematic Control Unit Module	Phone Module, Telematics module
TEL	Telephone Control Module	Bluetooth Phone Module
TPM	Tire Pressure Monitor Module	Tire Pressure Monitoring System (TPMS) Module
TRM	Trailer Module	Trailer Module
TSTR	Diagnostic Tester	
VDM	Vehicle Dynamics Control Module	Suspension Module
VSM	Vehicle Security Control Module	
YRM	Yaw Rate Sensor Module	Body Sensor Cluster

OVERVIEW

The Integrated Diagnostic System (IDS) is based on a Panasonic Toughbook® laptop computer which has been specifically designed to operate in a harsh environment. Combined with the Vehicle Communication Module (VCM) or a J2534 Vehicle Communication Interface (VCI), and either the Jaguar-specific Vehicle Measurement Module (VMM) or the Land Rover-specific T4 docking station, the system offers diagnostics for current and future Jaguar and Land Rover vehicles.

IDS includes a mobile workstation for docking and storing the laptop and associated hardware.

NOTE: Refer to the section 'Using T4 with IDS' for T4-specific information.

Current IDS Mobile Workstation



IDS SPECIFICATIONS

In the time since IDS was first introduced, PC specifications have advanced, so the software has been enhanced to take advantage of the hardware capabilities. At the time of printing, the current minimum specification is as follows:

- 1.06 GHz Dual Core Processor
- 1GB Ram
- 160GB Hard Drive

These specifications are equivalent to the Panasonic TOUGHBOOK® CF-19 Mk5.

Currently two models of approved Panasonic Toughbook® laptop computers are available: the CF-19 Mk5 and the CF-53 Mk1. Panasonic designed the Toughbook® with a magnesium case and spill-resistant keyboard to permit its use in workshop conditions.



IDS SOFTWARE

The IDS software provides the following diagnostic capabilities:

- OBD II diagnostics
- Full DTC read and Datalogger diagnostics
- Full vehicle configuration
- Module calibration update
- IDS diagnostic self-test

IDS software supports Jaguar and Land Rover vehicles as follows:

Jaguar

- 1995 model year onwards: All vehicles
 - All systems

Land Rover

- 2005 model year Range Rover
 - Infotainment systems
- 2005 model year LR3
 - All systems
- 2006 model year onwards: All vehicles
 - All systems

To ensure that all IDS units are configured correctly and consistently throughout the dealer network, IDS will be distributed with Microsoft® Windows® XP Service Pack 2 operating system already installed and IDS correctly configured.

NOTE: The IDS diagnostic application software must be loaded at the dealership.

 **CAUTION:** Do not attempt to update Microsoft® Windows® XP by downloading Microsoft® Windows® updates from the Microsoft® web site since these updates have not been validated with IDS. Failure to follow this instruction may prevent IDS from operating correctly.

Installing IDS Software

Dealers are required to install the IDS diagnostic software before IDS is ready for use. The latest SDD software is obtained at the IDS Central website at:

www.DiagnosticDelivery.com/IDSCentral

The diagnostic software is loaded from a flash drive onto the hard drive of the IDS laptop.

The installation procedure varies depending on the model of Toughbook® being used. Refer to the IDS/ SDD Maintenance Guide and any Read Me files on the IDS Central website, if necessary (details are provided in the Appendix).

Further details of the IDS software installation procedure are shown in the user guide, which is distributed with IDS.

Installing IDS Software Updates

Updated software releases are distributed to dealers over the internet. If software enhancements are required prior to the next software release, they are provided as automatic updates.

www.jlridssddmongoose.com

THIS PAGE LEFT BLANK INTENTIONALLY.

IDS-SDD-JLR: IDS Symptom Driven Diagnostics



Component Description



www.jlridssddmongoose.com

This publication is intended for instructional purposes only. Always refer to the appropriate service publication for specific details and procedures.

All rights reserved. All material contained herein is based on the latest information available at the time of publication. The right is reserved to make changes at any time without notice.

© 2012 Jaguar Land Rover North America LLC

<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

Overview	2
IDS Workstation	3
IDS Laptop	4
Vehicle Communication Module (VCM)	9
J2534 Vehicle Communication Interface Devices	12

OVERVIEW

The Integrated Diagnostic System (IDS) is a laptop PC-based diagnostic system using vehicle communication interfaces. An IDS workstation provides a docking station for the laptop, allowing for power supply, battery charging, and secure locking. It also provides storage for the remaining components that make up the system.

Currently two models of approved Panasonic Toughbook® laptop computers are available: the CF-19 Mk5 and the CF-53 Mk1. Panasonic designed the Toughbook® with a magnesium case and spill-resistant keyboard to permit its use in workshop conditions.

In the time since IDS was first introduced, PC specifications have advanced, so the software has been enhanced to take advantage of the hardware capabilities. At the time of printing, the current minimum specification is as follows:

- 1.06 GHz Dual Core Processor
- 1GB Ram
- 160GB Hard Drive

These specifications are equivalent to the Panasonic TOUGHBOOK® CF-19 Mk2.

Additional Hardware

Additional IDS hardware at your dealership may include the following:

- Vehicle Communication Module (VCM) or J2534 Vehicle Communication Interface (VCI)
- Vehicle Measurement Module (VMM) (Jaguar only)
- RDS T4 docking station (Land Rover only)
- Appropriate test leads and accessories
- External DVD/CD-RW drive (no longer used for diagnostics, but may be used for Navigation map updates)

Accessory socket power supply (optional)

Removable USB flash drive

IDS WORKSTATION

The IDS workstation cart provides a docking station and storage for the IDS laptop and associated hardware.

The workstation includes an AC power connection with a built-in transformer to provide the required DC voltage for the laptop when docked, and allows the laptop to be charged from an electrical outlet. The later style cart provides the option of adding a separate flat-screen monitor.

Early Style Cart



Later Style Cart



IDS LAPTOP

The Panasonic Toughbook® IDS laptop is the primary component of the IDS system, and can be used docked to the workstation or undocked as a portable unit.

NOTE: The laptop uses Local Area Network (LAN) wireless to receive diagnostic software downloads and updates. The wireless antennas are located in the casing of the lid around the display.

Both the CF-19 and CF-53 have Touch-Screen Displays (TSD) for easy operation of the diagnostic software. The CF-19 Toughbook is designed so that it can be used in standard Laptop Mode or with the screen rotated and flattened to the unit in Tablet Mode.



PANASONIC CF-19 Mk5

SDD-JLR254



PANASONIC CF-53 Mk1

Toughbook Specifications

Panasonic CF-19 Mk5	Panasonic CF-53 Mk1
- Fully-Rugged TOUGHBOOK® - Windows 7 - Intel Core I5-2520M 2.50 GHz processor 10.4" HD Touchscreen LCD 320GB Shock Mounted HD (7200 RPM) 4GB Memory - Wi Fi - Bluetooth®	- Semi-Rugged TOUGHBOOK® - Windows 7 - Intel Core I5-2520M 2.50 GHz processor 14.0" HD Touchscreen LCD 320GB Shock Mounted HD (7200 RPM) 4GB Memory - Wi Fi - Bluetooth® - DVD Super Multi Drive

IDS Laptop Features

CF-19 Mk5



Power Switch

The power switch is located at the right front of the CF-19 laptop.

Battery and Power Status LEDs

The battery and power status LEDs are located along the right front edge of the laptop.

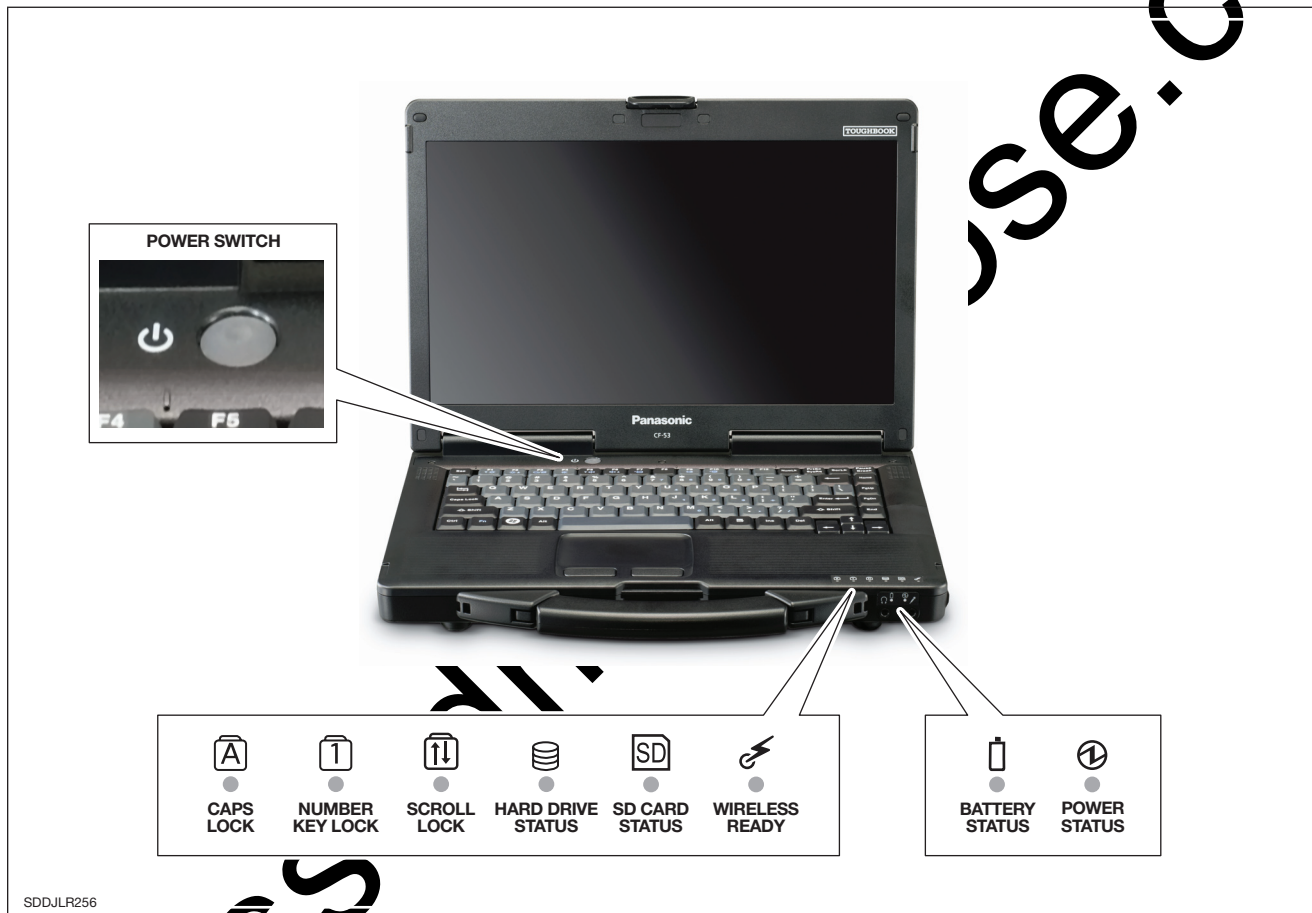
Power Status

- Off: Power Off / Hibernating
- Green: Power On

LED Indicators

The LED status indicators are located along the left front edge of the laptop.

CF-53 Mk1

**Power Switch**

The power switch is located at the rear of the keyboard below the display.

Battery and Power Status LEDs

The battery and power status LEDs are located at the right front of the laptop.

Power Status

- Off: Power Off / Hibernating
- Green: Power On
- Blinking Green: Sleep / Standby
- Rapid Blinking Green: Cannot power up due to low temperature

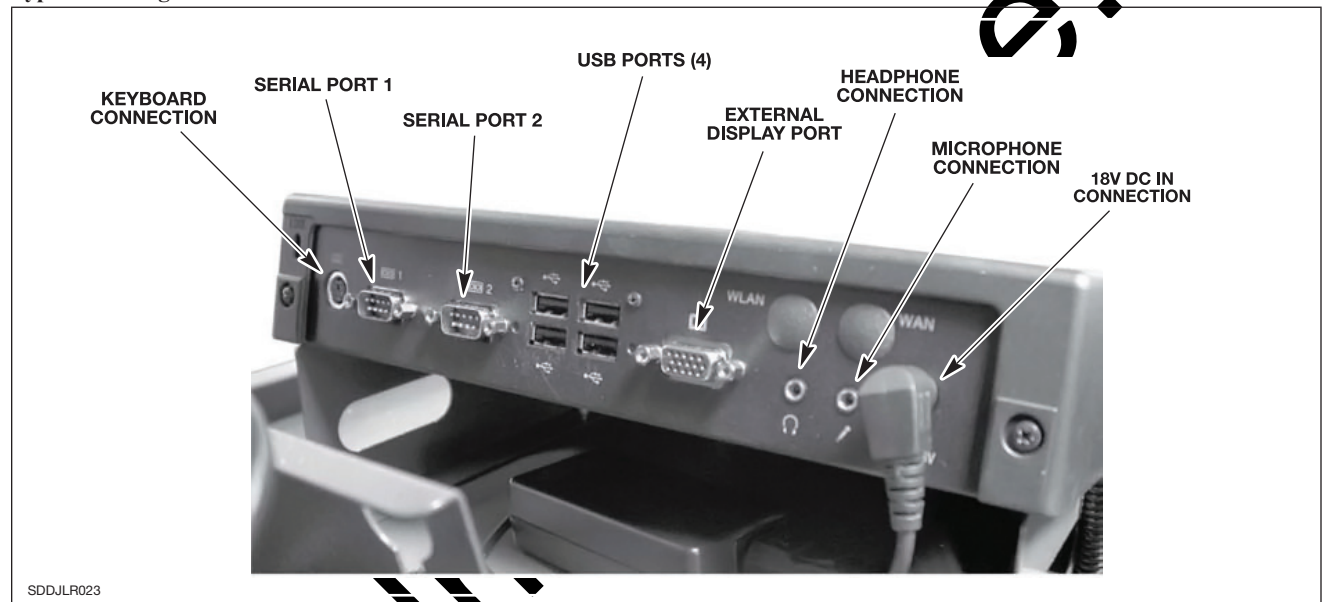
LED Indicators

The LED status indicators are located along the right front edge of the laptop.

Docking Station Connections

When the laptop is docked to the workstation, most of the laptop ports are replicated on the rear of the docking station.

Typical Docking Station Connections



The USB ports are used to connect the following components:

- VCM
- VMM
- DVD / CD / RW drive

The fourth USB port may be used in the future.

Laptop Power Supply

IDS can be powered from an AC electrical supply through a DC transformer.

The IDS workstation has its own built-in transformer to convert AC power to the required DC voltage for IDS. When the IDS is docked to the workstation it is automatically connected to the AC power supply transformer.

If the laptop is removed from the workstation, a separate AC power supply transformer (supplied with the workstation) can be connected to it using the power supply socket located at the left hand side of the laptop computer.

Separate AC Power Supply Transformer

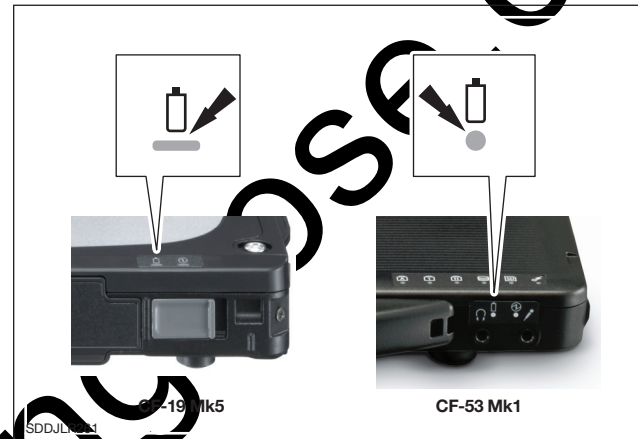


Toughbook Internal Battery

The IDS laptop also can be powered by its internal Lithium battery, which is located at the right hand side of the laptop.



Battery Status LEDs



LED Indicator	Battery Status
Not Lit	The battery pack is not inserted or is not being charged.
Orange	Charging is in progress.
Green	The battery is fully charged.
Blinking Green	In High Temperature Mode; the battery pack discharges the power to 80% of the normal temperature mode. Do not remove the battery pack in this status.
Red	The remaining battery charge is approximately 9% or less.
Blinking Red	The battery pack or the charging circuit is not operating properly.
Blinking Orange	The battery cannot be charged (temporarily) for one of the following reasons: - Its internal temperature is out of the acceptable range. - The power supply is insufficient because software applications or peripheral devices are consuming a large amount of power.

Battery Charging Best Practices

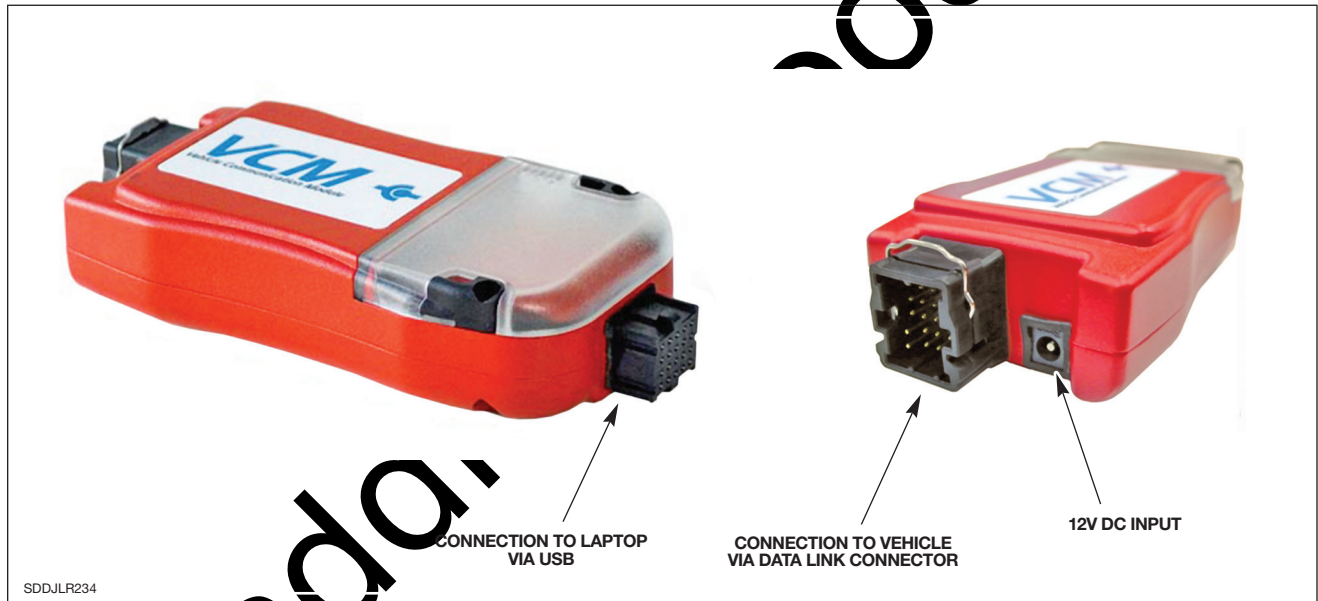
- The battery life deteriorates each time it is charged. Charging the battery only when it has become completely discharged will reduce the number of charge cycles and thus extend battery life.
- Do not repeatedly charge the battery before it is discharged.
- To help reduce the number of charge cycles, the battery can only be recharged when the remaining power is less than 95% of its capacity.
- The battery will not charge when it is outside its temperature range of 10°C to 30°C, (50° to 86°F). If the battery is outside of this temperature range, the battery indicator lamp will flash orange and charging will not begin.

Battery Calibration

Over time, the remaining battery capacity may not be displayed accurately. In this situation, it may be necessary to carry out the battery recalibration procedure. Additional information on the battery calibration procedure is provided in the 'Appendix' section of this course.

VEHICLE COMMUNICATION MODULE (VCM)

The Vehicle Communication Module (VCM) is a high-performance vehicle serial communications gateway which facilitates communication between IDS and the various vehicle networks (such as CAN, SAE and ISO9141) of most current and future of Jaguar/Land Rover vehicles. The VCM also allows communication with most legacy vehicles (those prior to the introduction of ISO14229 diagnostic protocol). The VCM is a rugged unit housed in a magnesium casing with a tough protective plastic cover.



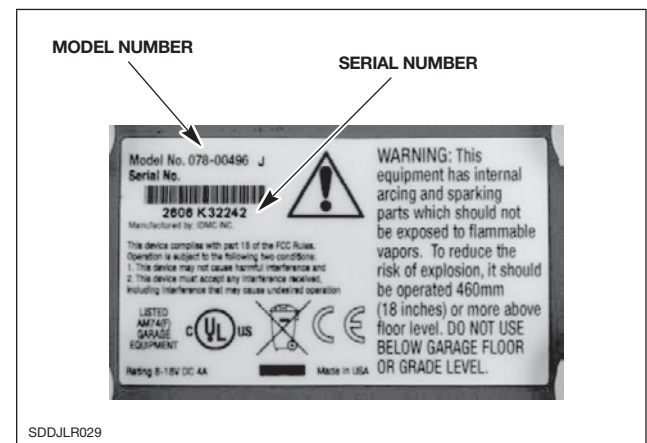
Genuine VCMs are currently only manufactured in the USA.

The VCM provides three sockets for external connections. Two of the sockets allow connection to IDS and to the vehicle. The third socket is to allow for a stand-alone 12V DC power supply, which is not currently provided with Jaguar applications. The VCM gets its power from the vehicle via the data link connector, NOT from the laptop.

NOTE: The VCM power supply, which is available from SPX, will not charge the IDS laptop battery when connected.

CAUTION: Excessive stress on the rear USB connector can damage the USB cable, disrupting VCM communication. Care must be taken when installing or removing the VCM to route the USB cables with minimal stress. The cables are long to facilitate remote use; be sure to route excess length such that the cable will not be pinched by the workstation enclosure.

The VCM serial and model number can be found on a label on the underside of the unit. These numbers may be required when requesting IDS support.



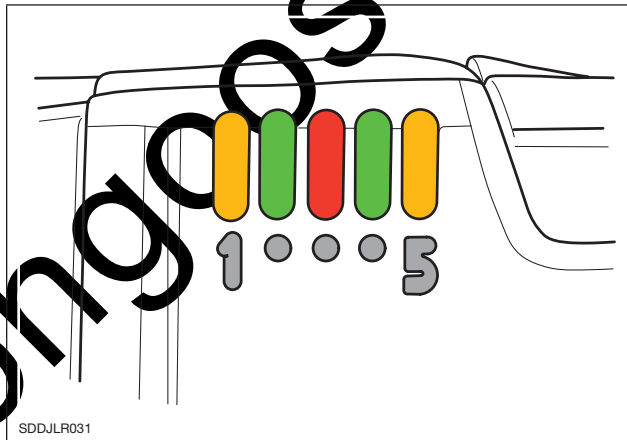
The VCM is located behind the laptop docking area of the IDS workstation, attached to the bracket of the workstation by locating lugs sliding into 'key holes' on the bracket. The VCM is easily removed from the bracket when required.

VCM Location



VCM LED Indicators

The VCM features five status indicator LEDs, visible through the plastic cover of the VCM to allow the user to visually observe the operation of the VCM. LEDs are numbered 1 – 5 from left to right.

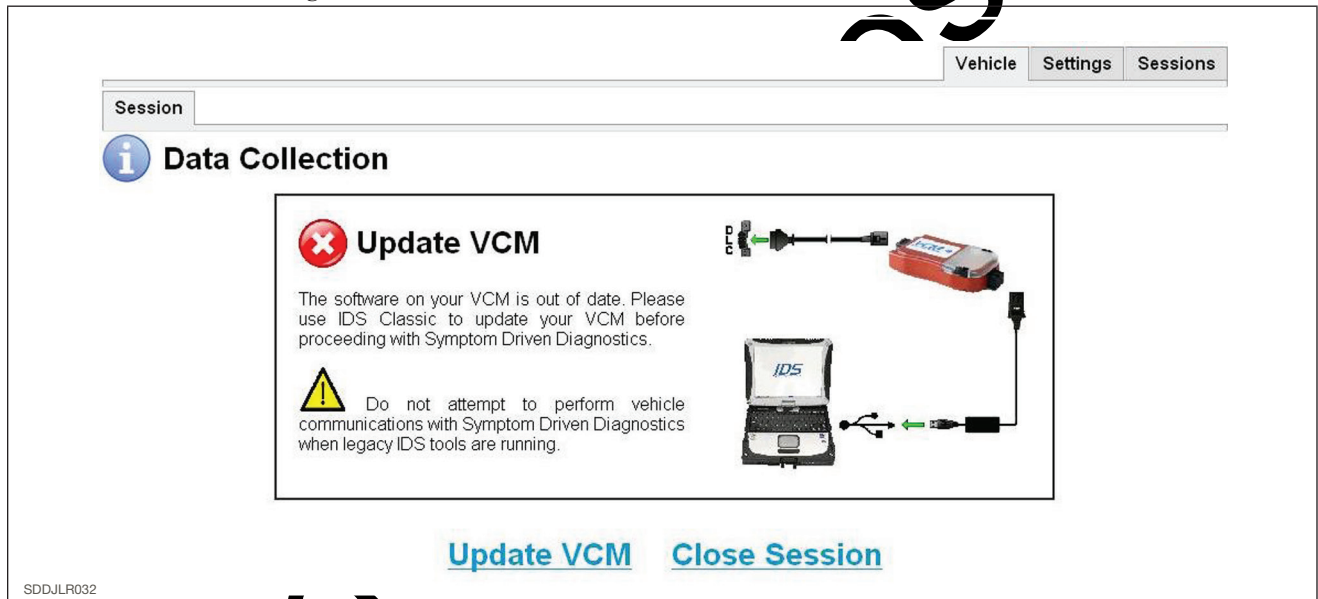


LED	Color	Indicates	Operating Behavior
1	Amber	Vehicle link established	Flashes when communication link to vehicle is established
2	Green	VCM operating	Flashes @ regular interval when functioning
3	Red	Power supply	Lit when functioning
4	Green	Flash memory access	Should be OFF except during VCM software update
5	Amber	Host Link	Flashes when communication link to Toughbook is established

VCM Driver Software

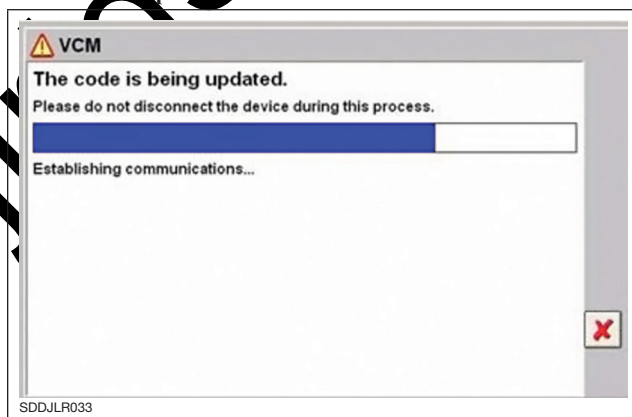
Before the VCM can be used, it is necessary to download the latest driver software. Each time the VCM is connected to IDS and the VCM is powered up, the VCM driver software version will be checked. If the current software of the VCM is out of date and a later version is available, a message will be displayed stating that the code is out of date and asking if the user wants to update the VCM now.

VCM Code Out of Date Message



NOTE: Always select the affirmative tick (check mark) in this case. A status bar screen will be displayed while the driver is updated.

VCM Driver Update Status



Downloading the latest VCM driver software will only take a short time (approximately 90 seconds to complete). Once the software has been downloaded, diagnostics with the vehicle may be carried out as normal.

CAUTION: Do not switch off IDS or disconnect the VCM during a software download. Failure to follow this instruction may cause damage to the VCM or may cause communication errors when attempting to diagnose vehicle faults.

NOTE: A VCM software download is only initiated when a later software version is available following an IDS software update. Later VCM software cannot be overwritten by an earlier software version.

NOTE: The VCM recovery procedure is described in the IDS/SDD Maintenance Guide.

J2534 VEHICLE COMMUNICATION INTERFACE DEVICES

Starting with IDS DVD 125 (March 2011), the IDS/SDD diagnostic platform supports a number of Jaguar Land Rover-approved J2534 Vehicle Communication Interface (VCI) devices in addition to the VCM.

J2534 is an international standard for vehicle communication devices known as ‘pass-thru’ devices. Pass-thru devices allow a PC-based system to communicate with and/or program modules on supported vehicles. This

capability allows the user to read DTCs, monitor PIDs, and reprogram Control Modules. Previously, the only approved communication device for use with IDS/SDD has been the Teradyne Vehicle Communications Module (VCM).

Support for these devices is specified by the EU5 standard. For Jaguar and Land Rover vehicles, this represents 2011MY NA5 vehicles and later.

Model Part	Number
Diagnostic Associates DA-VINA	JLR-DAVIN0001
Diagnostic Associates DA-Dongle	JLR-DAVIN0021
Drewtech Mongoose	JLR-MONGOOSE-VCI
Teradyne iView	JLR-iVIEW-VCI

NOTE: As required under EU5 vehicle compliance legislation, the IDS/SDD Diagnostic System is capable of supporting additional VCI devices. However, these units will not be supported by Jaguar Land Rover Technical Support or SPX Customer Service Help Line, and will not be listed on the Jaguar Land Rover approved list.

Diagnostic Associates DA-VINA



Drewtech Mongoose



Diagnostic Associates DA-Dongle



Teradyne iView



www.jlridssddmongoose.com

THIS PAGE LEFT BLANK INTENTIONALLY.

IDS-SDD-JLR: IDS Symptom Driven Diagnostics



Introduction to Symptom Driven Diagnostics



www.jlridssddmongoose.com

This publication is intended for instructional purposes only. Always refer to the appropriate service publication for specific details and procedures.

All rights reserved. All material contained herein is based on the latest information available at the time of publication. The right is reserved to make changes at any time without notice.

© 2012 Jaguar Land Rover North America LLC

<https://www.jlridssddmongoose.com/sdd-symptom-driven-diagnostics/>

Overview	2
System Settings	8
Working with SDD	14
Pre Delivery Inspection (PDI) Session Type	22
Measurement Applications Session Type	27
Service Functions Session Type	35
Campaigns Session Type	37
Diagnosis Session Type	40
Sample Diagnostic Routine	46
Vehicle Sessions	67
Repair Validation	72
Initial Data Reinstatement	73
Events Definitions	75

OVERVIEW

Why Change IDS?

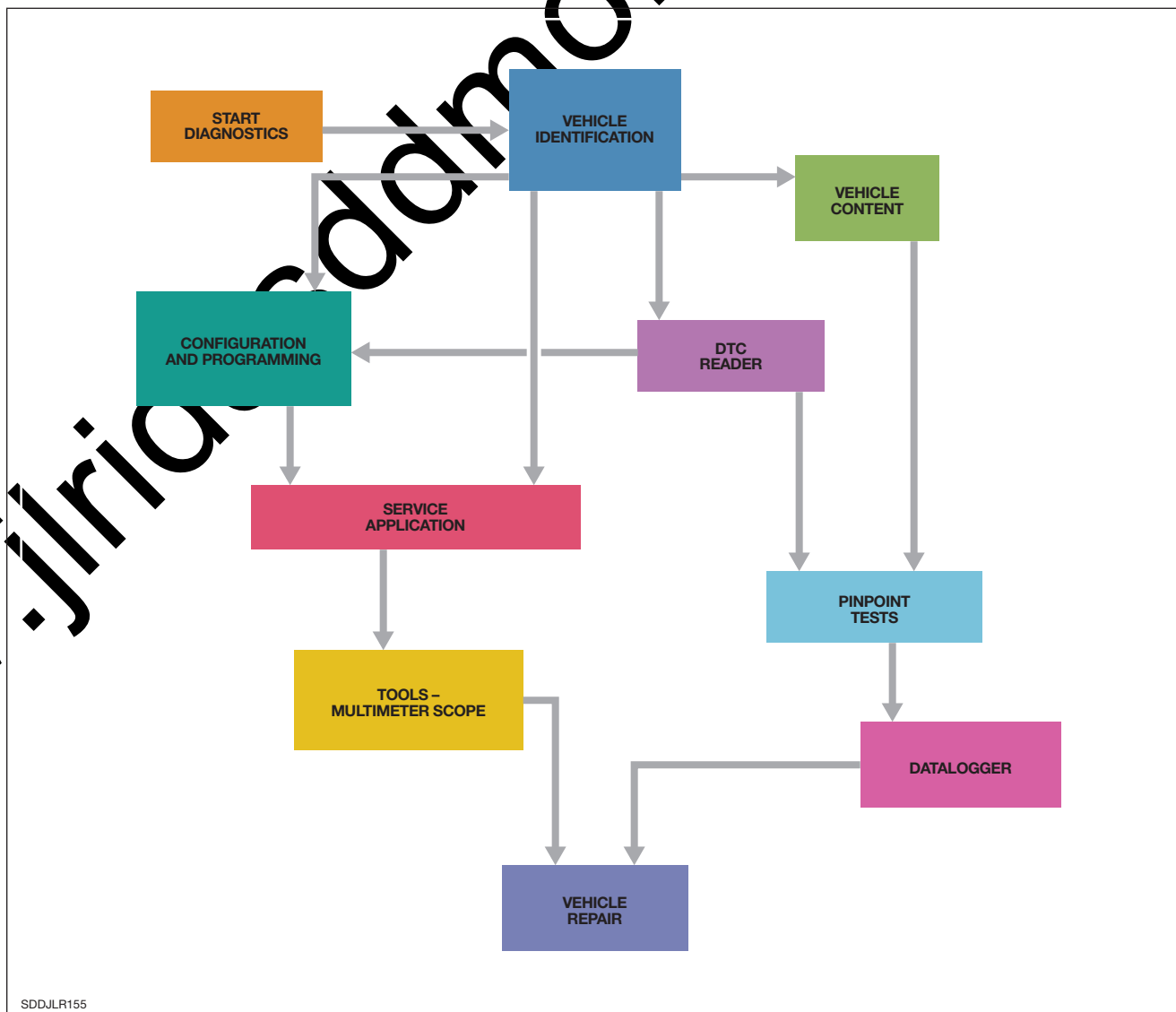
The Integrated Diagnostic System (IDS) core architecture dates from 1999 and incorporates elements going back as far as 1995. In the meantime, vehicle technology has become more complex and the number of vehicle variants has increased. Current vehicle systems support up to 4,000 different Diagnostic Trouble Codes (DTCs), which increases the complexity of diagnostics. The technician now needs to diagnose systems, not merely individual components.

Without a clear strategy for a repair process, a technician faced with up to 40 DTCs at the start of an IDS session can easily lose confidence in the diagnostic tool. Feedback data from dealerships shows inconsistent methods of fault diagnosis with under-utilization of the IDS capabilities.

This has resulted in high 'No Fault Found' repair rates, difficulty in achieving Fixed Right First Time (FRFT) repairs, and an increase in the time that a customer's vehicle is off the road.

Unstructured Diagnosis

The diagram shows a typical flow chart of a diagnostic procedure with little or no clear direction of a systematic process.



New Diagnostic Process

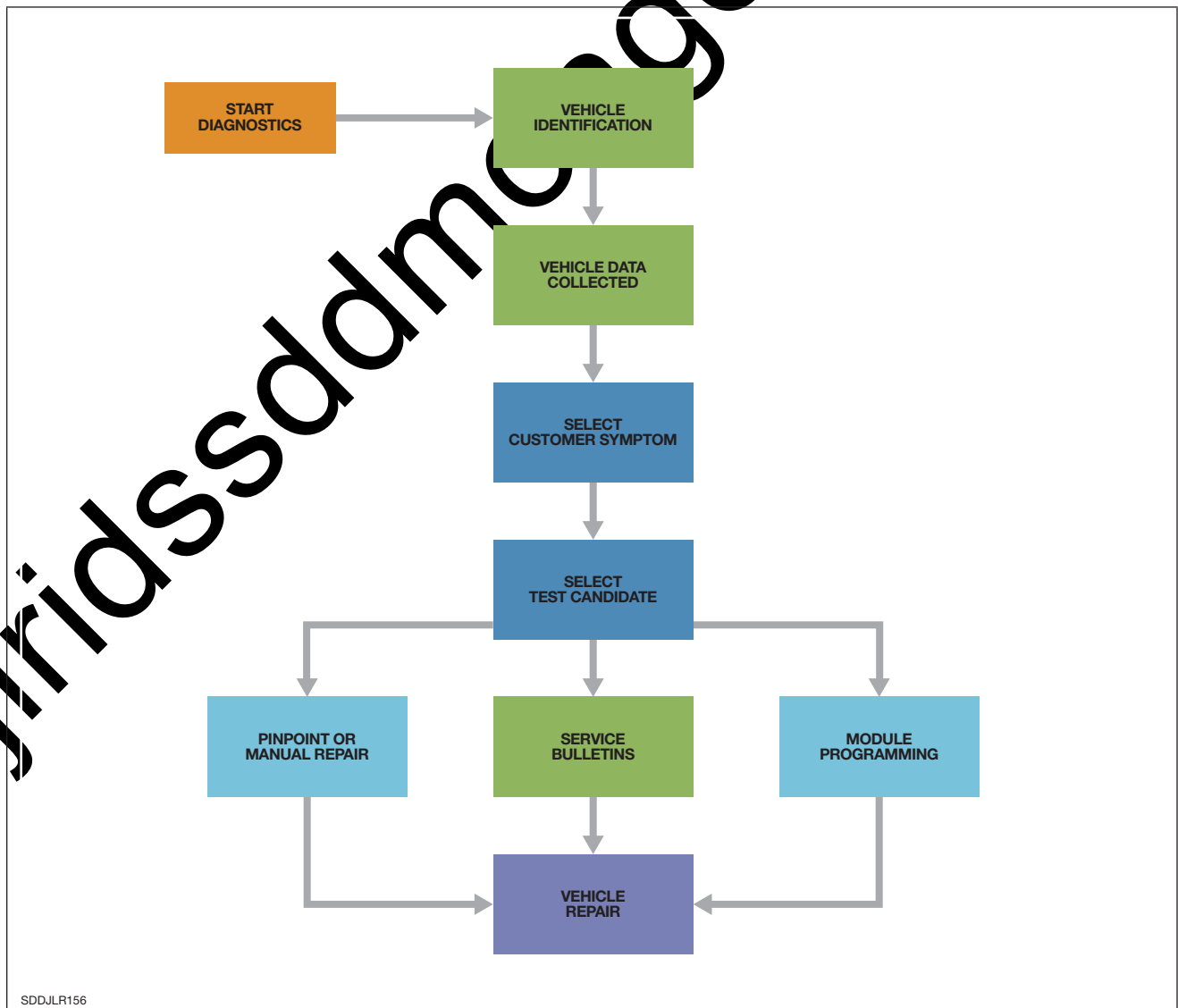
IDS has been updated with the goal of providing a clear diagnostic strategy and tool navigation path to the technician. With Symptom Driven Diagnostics (SDD), the technician is able to begin the diagnostic process by identifying the specific symptoms of a customer concern using Symptom Maps incorporated into the

SDD software. A diagnostic strategy can then be structured around only the DTCs that are relevant to those symptoms.

SDD can also provide links to other service information to support the diagnostic process.

Structured Diagnosis

The diagram shows a schematic of the systematic application of a structured diagnostic route using symptom-based information and diagnostic Symptom Maps.



Symptom Maps

Symptom Maps are incorporated into the SDD software, and allow the technician to target DTCs and diagnostic routines based on a specific set of symptoms. Symptom Maps are organized by major vehicle system as follows:

- Module Communications Network
- Chassis
- Powertrain
- Electrical
- Body

Maps are then further structured by System and Sub-System details to provide lists of possible symptoms at 2 levels.

Symptom Maps are used by SDD in conjunction with DTCs. In order to ensure the best possible recommendations to achieve FRFT repairs, the technician must input ALL vehicle symptoms that match the customer concern.

NOTE: Symptom Maps are viewed from the Symptom Selection Screen as described later in this section. Symptom Maps can also be searched using the browser's 'Find' feature (Ctrl F).

Use this, Base Vehicle System, level to navigate to Detailed System and Symptom options, that are currently supported for Chassis faults. Below is a Chassis Symptom Map table that identifies many of the Detailed System, Sub-system and Symptom options currently supported for the Chassis System. For additional information relating to these Systems and Sub-systems, please refer to the relevant Description and Operation, and Diagnosis and Testing Sections in the Workshop Manual.

Chassis Symptom Map

Base Vehicle System	Detailed System	Detailed Sub-system	Symptom (Level 1)	Symptom (Level 2)
Chassis	Suspension System	Mechanical Suspension	Height	Too High
				Too Low
			Too Harsh	
			Bounces and Rebounds	
			Noises	
			Leaks	
		Vehicle Dynamic Suspension	Height	Adjustment Inoperative Using Remote Key Handset
				Adjustment Inoperative Using Manual Switch
				Automatic Ride Leveling Adjustment Inoperative
				Too High
				Too Low
				Uneven
				Restricted to Default Safety Mode
				Delayed Operation
	Constantly Changes Up and Down			

SDD JLR157

IMPORTANT NOTE: Because SDD relies on logged DTCs in conjunction with vehicle symptoms to perform diagnostics, it is very important that you **DO NOT CLEAR DTCs** unless instructed to do so by SDD. Clearing DTCs will clear the vehicle fault history and erase important data required for correct diagnosis of a concern.

Sample Symptom Map

The table below approximates the Symptom Map for the Powertrain System as currently included in IDS. Symptom Maps will continue to evolve, and will be updated with subsequent software releases.

Basic Vehicle System (Level 1)	Detailed System (Level 2)	Detailed Sub-system (Level 3)	Symptom (Level 4)	Symptom (Level 5)
Powertrain	Engine System	Engine Cooling	Engine Overheats	
			Engine Will Not Reach Operating Temperature	
			Coolant Loss	
			Cooling Fan Inoperative	
			Cooling Fan Inoperative	
		Starting System	Will Not Crank	
			Will Not Start	
				Cranks Too Slow
				Cranking Speed Normal
				Engine Cranks Without Switch Input
		Starting System	Starts With Incorrect Starting Conditions	Engine Cranks With Incorrect Switch Input
				Starts Without Operating Brake Pedal
				Starts Without Operating Clutch Pedal
				Smart Cranking Will Not Operate
				Delay Before Cranking
		Fuel Vapor and Odor	Inside Vehicle	
			Outside of Vehicle	
		Fuel Tank	Slow Fill and Splashback	Leaks
			Leaks	
		Emissions	Excessive Smoke From Exhaust	
			Odor From Exhaust	
			Exhaust Noises	Rattle
				Harmonics Audible at Low Speed
				Harmonics Missing Under Acceleration
			Secondary Air Injection	Air Pump Runs Continuously
		Glow Plug System	Inoperative	
			On Constantly When Ignition On	
			Warning Lamp Will Not Illuminate	

Basic Vehicle System (Level 1)	Detailed System (Level 2)	Detailed Sub-system (Level 3)	Symptom (Level 4)	Symptom (Level 5)
Powertrain	Engine System	Engine Performance	Vehicle In Limp Home Mode	
			Fuel Consumption High	
			Poor Idle	
			High Idle	
			Poor Acceleration and Lack of Power	
			Engine Hesitates	
			Engine Vibrates	
			Engine Surges	
			Engine Judders and Shakes	
			Engine Misfire	
			Engine Stalls	
			Noises	
		Speed Control System	Adaptive Speed Control	Adaptive Speed Control Inoperative
			Speed Control	Speed Control Does Not Release
				Speed Control Inoperative
			Automatic Speed Limiter	Inoperative
	Automatic Transmission and Transaxle	Automatic Transmission and Transaxle Symptoms	Transmission In Limp Home	
			Holds Gear for Too Long	
			Harsh Gear Engagement From Park or Neutral	
			Harsh Gear Change	
			Will Not Change Up Gear	
			Will Not Change Down Gear	
			Winter Mode Inoperative	
			Sport Mode Permanently Selected	
			Torque Converter Clutch Inoperative	
			Slips in Gear	
			Fluid Leaks	
			Kickdown Inoperative	
			Judder	
			Noises	
			Bump From Transmission On Shutdown	
			Transfer Case Inoperative	
			High or Low Range Selection Switch Inoperative	
			Dynamic Optimization Switch Inoperative	

Basic Vehicle System (Level 1)	Detailed System (Level 2)	Detailed Sub-system (Level 3)	Symptom (Level 4)	Symptom (Level 5)
Powertrain	Automatic Transmission and Transaxle	Automatic Transmission and Transaxle Symptoms	Gear Selection Malfunction	Selector Locked in Park
				Selector Inhibit Inoperative
				Selector Position Locked
				Selector Reduced Operation
				Selector Does Not Raise or Lower
				Selector Will Not Lock at Ignition Off
				Gear Position Display Flashing
				Reverse Gear Inoperative
				No Drive In Any Selected Position
				Step Shift Inoperative
				Inconsistent Operation
				Noisy
				Coarse Operation
				Display Position Illumination Fault
				Does Not Upshift
				Inoperative
				Does Not Downshift

NOTE: Additional information on Systems and Sub-systems can be found in the Workshop Manual 'Description and Operation' and 'Diagnosis and Testing' Sections.

Recommendations

Based on the symptoms selected and the DTCs present in the vehicle modules, SDD will provide the technician with a list of recommendations for diagnosis and repair, which may include the following:

- Pinpoint Test
- Technical Service Bulletin
- Functional Test
- Service Function
- Diagnostic Tools

Recommendations are identified by icons as shown in the graphic.

